School of Future Tech

Case Study Report

on

Multi-Campus University Management System

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Introduction to the Case Study

A university with multiple campuses needs an efficient way to manage student information from different locations. Managing this data manually or separately for each campus can be difficult and time-consuming.

This case study focuses on building a simple multi-campus university management system using C++. The system allows administrators to manage student records from different campuses in one centralized program.

The application performs operations such as the following:

- Adding new student records
- Viewing all student information
- Searching students by name
- Filtering students by campus
- Updating student information
- Deleting student records
- Generating a central report of all campuses

The system stores data in a file, which allows information to be saved permanently even after the program is closed.

Case Background

Managing student records across multiple university campuses is complex when each campus maintains separate records. Without a centralized system, it becomes difficult to track student data, generate reports, or maintain accurate

information. The purpose of this project is to design a simple ERP-style management system that can centrally manage student data from different campuses. The system provides a menu-based console application that allows users to manage and access student records efficiently. The data is stored in a text file so that records can be reused whenever the program is executed again.This solution demonstrates how basic programming concepts, data structures, and file handling in C++ can be used to solve real-world management problems.

Case Study Design

The system is designed as a menu-driven console application.

The main components of the system include:

Student Structure

A structure is used to store student information.

Fields include:

- Student ID
- Student Name
- Campus Name
- Course Name
- Fees

Data Storage

Student records are stored using a vector, which acts as a dynamic list.

```
vector<Student> students;
```

File Storage

All student data is saved to a file named

students.txt

This ensures that records are preserved even after the program exits.

System Modules

The system contains the following functional modules:

1. Add Student
2. View All Students
3. View Students by Campus
4. Search Student by Name
5. Update Student Information
6. Delete Student
7. Central Report Generation
8. Exit System

These modules help manage student records efficiently.

Methods & Algorithms Technology Applied

Several programming concepts and technologies are used to implement the system.

1. C++ Programming Language

The entire system is developed using C++, which provides powerful features for structured programming.

2. Data Structures

The program uses the vector container to store student records dynamically.

```
vector<Student> students;
```

Vectors allow flexible data storage without fixed memory limits.

3. File Handling

The program uses the fstream library to read and write data.

```
ofstream file("students.txt");
ifstream file("students.txt");
```

This allows data persistence.

4. Searching Algorithm

A linear search algorithm is used to find students by name or ID.

```
Example: for (const Student& s : students)
```

5. Data Encoding and Decoding

Special functions are used to avoid errors when saving data with commas.

```
encode()
decode()
```

These functions replace commas with another character before saving.

6. Map Data Structure

The map container is used to count the number of students in each campus.

```
map<string, int> campusCount;
```

Case Study Implementation Details and Snapshots

The Multi-Campus University Management System is implemented using the C++ programming language as a menu-driven console application. The system allows users to manage student information from multiple campuses in a centralized way.

A structure named "Student" is used to store student details such as student ID, name, campus, course, and fees. All student records are stored in a vector, which allows dynamic storage and easy management of multiple records.

The program also uses file handling to save and load student data from a file named students.txt. This ensures that the data remains stored even after the program is closed.

The system provides several functionalities, such as:

- Adding a new student
- Viewing all student records
- Viewing students by campus
- Searching students by name
- Updating student information
- Deleting student records
- Generating a central report of students and fees

A map data structure is used in the reporting module to count the number of students from each campus. The program also includes input validation to ensure correct data entry.

Main Menu Screenshot

Show the menu when the program starts.

```
cd "/Users/ishitanakhawa/C++ case study/" && g++ university_erp.cpp -o university_erp && "/Users/ishitanakhawa/C++ case study/"university_erp
(base) ishitanakhawa@Ishitas-MacBook-Air C++ case study % cd "/Users/ishitanakhawa/C++ case study/" && g++ university_erp.cpp -o university_erp &
& "/Users/ishitanakhawa/C++ case study/"university_erp

====== Multi-Campus University ERP ======
  1. Add Student
  2. View All Students
  3. View Students by Campus
  4. Search Student by Name
  5. Update Student
  6. Delete Student
  7. Central Report
  8. Exit
========================================
  Enter Choice:
```

Add Student

This function collects student details and saves them.

```
Enter Choice: 1
Enter Name   : Aisha
Enter Campus : ITM,Delhi
Enter Course : BTech-CSE
Enter Fees   : 2000000.00

Student added! ID is: 1
```

View All Students

Displays all students stored in the system.

```
====== Multi-Campus University ERP ======
  1. Add Student
  2. View All Students
  3. View Students by Campus
  4. Search Student by Name
  5. Update Student
  6. Delete Student
  7. Central Report
  8. Exit
==========================================
  Enter Choice: 2

======== All Students ========
  ID     : 1
  Name   : Aisha
  Campus : ITM,Delhi
  Course : BTech-CSE
  Fees   : 2000000.00
  --------------------------------
  ID     : 2
  Name   : Anna
  Campus : ITM,Mumbai
  Course : BTech-CSE
  Fees   : 2000000.00
  --------------------------------
  ID     : 3
  Name   : Jack
  Campus : ITM,Gujarat
  Course : BTech-CSE
  Fees   : 2000000.00
  --------------------------------
```

Search Student by Name

Allows searching using keywords.

```
====== Multi-Campus University ERP ======
  1. Add Student
  2. View All Students
  3. View Students by Campus
  4. Search Student by Name
  5. Update Student
  6. Delete Student
  7. Central Report
  8. Exit
==========================================
  Enter Choice: 4
  Enter Name to Search: Anna

===== Search Results =====
  ID     : 2
  Name   : Anna
  Campus : ITM,Mumbai
  Course : BTech-CSE
  Fees   : 2000000.00
  --------------------------------
```

Update Student

Allows modifying student details using the student ID.

```
====== Multi-Campus University ERP ======
  1. Add Student
  2. View All Students
  3. View Students by Campus
  4. Search Student by Name
  5. Update Student
  6. Delete Student
  7. Central Report
  8. Exit
==========================================
  Enter Choice: 5
  Enter Student ID to Update: 3
  Enter New Name   : Rohan
  Enter New Campus : ITM,Chennai
  Enter New Course : BTech-CSE
  Enter New Fees   : 2000000.00

  Student updated successfully!
```

Delete Student

Removes a student record after confirmation.

```
====== Multi-Campus University ERP ======
  1. Add Student
  2. View All Students
  3. View Students by Campus
  4. Search Student by Name
  5. Update Student
  6. Delete Student
  7. Central Report
  8. Exit
==========================================
  Enter Choice: 6
  Enter Student ID to Delete: 4
  Delete 'Jack'? (y/n): y

  Student deleted successfully!
```

Central Report

Generates a summary of students and total fees.

```
====== Multi-Campus University ERP ======
  1. Add Student
  2. View All Students
  3. View Students by Campus
  4. Search Student by Name
  5. Update Student
  6. Delete Student
  7. Central Report
  8. Exit
==========================================
  Enter Choice: 7

======== Central Report ========
Total Students : 3
Total Fees     : 6000000.00

Students Per Campus:
  ITM,Chennai : 1
  ITM,Delhi : 1
  ITM,Mumbai : 1
================================
```

Case Study Results and Conclusion

The Multi-Campus University Management System successfully demonstrates how a centralized system can simplify student data management.

The program allows users to perform essential operations such as adding, updating, searching, and deleting student records. It also generates useful reports that summarize information across different campuses.

By using C++ vectors, file handling, and data structures, the system provides a simple yet effective solution for managing university records.

In conclusion, the project shows that even a console-based application can serve as a basic ERP-like system for managing multi-campus student data efficiently.

References

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